

From AI Experiments to Institutional Capability

A Stewardship Agenda for Public Pension Boards
and Executive Leaders

GOVERNING QUESTION How can a public pension system turn AI possibility into durable public value without outrunning trust?

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THE DANG FRAMEWORK | PUBLIC THESIS

Institutional wisdom for the technology age.

AUTHOR'S NOTE

About This Paper

This white paper extends the public conversation begun in the July 30, 2026 NCPERS webinar, “From Experimentation to Enterprise: Building an AI Strategy for Public Pension Systems,” presented by Darren Dang and OCERS Chief Executive Officer Steve Delaney. The webinar addressed high-value use cases, governance, organizational risk, workforce readiness, and responsible adoption in public pension organizations.

The paper offers a public leadership lens—not a proprietary implementation manual. It shares the principles and high-level architecture needed to improve institutional judgment while protecting detailed scoring, weights, thresholds, diagnostic instruments, playbooks, and institution-specific methods.

PUBLIC THESIS. PRIVATE METHOD. Teach the logic generously. Protect implementation selectively.

About the author

Darren Dang is a public-sector Chief Technology Officer whose work connects technology strategy, data, cybersecurity, modernization, artificial intelligence, governance, and organizational capability. His governing purpose is simple: make technology a force that elevates people and mission.

Independence and scope

The views expressed are the author's own and do not necessarily represent the official position of OCERS, NCPERS, or any other institution. This paper is educational and strategic in nature; it is not legal, investment, cybersecurity, procurement, or regulatory advice. Public pension systems should apply their own laws, policies, fiduciary duties, labor obligations, contracts, risk tolerances, and governance structures.

THE ANSWER FIRST

Executive Summary

Public institutions do not need more AI activity. They need the institutional capability to decide what to explore, what to stop, what to scale, and when to rebalance—and to explain those decisions with evidence.

An experiment can demonstrate possibility. It cannot, by itself, establish value, accountability, readiness, or trust. A successful pilot may still depend on fragile data, unclear ownership, vendor promises, informal workarounds, or a few unusually capable people. Scaling those conditions does not create institutional capability. It scales uncertainty.

For public pension systems, the standard must be higher. These institutions operate across generations. They steward sensitive information, essential services, public resources, and promises that members may rely on for decades. The question is not whether an AI tool is impressive. The question is whether its use makes the institution more capable—and more worthy of trust.

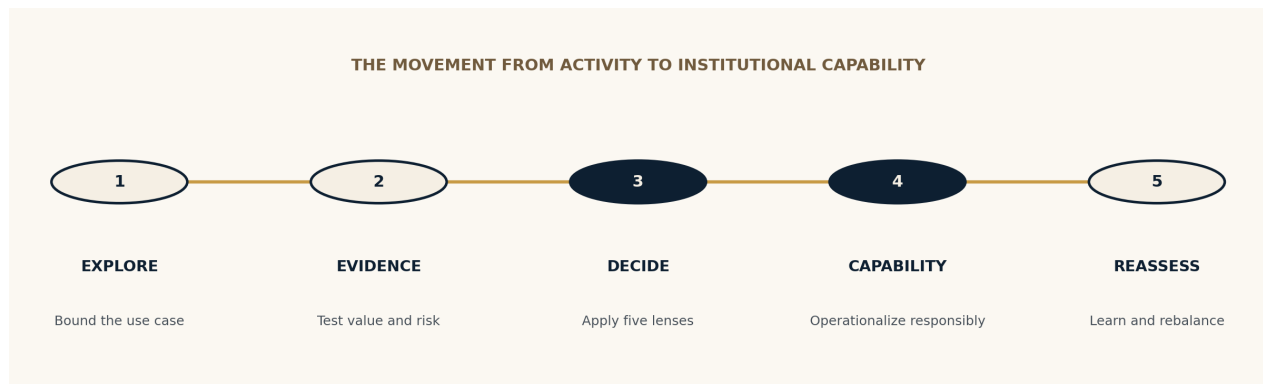


FIGURE 1 • Capability is a repeatable institutional discipline, not a larger collection of pilots.

This paper proposes five public questions before scale:

1. **Mission.** Does the use materially advance institutional purpose or member value?
2. **Value.** What measurable organizational return should it create?
3. **Trust.** How will it affect confidence, fairness, transparency, reliability, and responsible use?
4. **Risk.** What could fail, who could be harmed, and what exposure would remain?
5. **Readiness.** Can the people, processes, data, governance, architecture, capacity, and ownership support responsible execution?

Together, these questions form the public logic of the Dang Decision Test. They do not replace judgment. They discipline it.

The final measure is dual return: Organizational Return and Trust Return. Did the investment improve service, capability, productivity, resilience, or decision quality? Did it also strengthen accountability, reliability, fairness, transparency, and public confidence?

MISSION BEFORE TECHNOLOGY

1. The Institutional Question

Artificial intelligence is moving from isolated experimentation into the operating fabric of public institutions. California is testing generative AI in call-center research, language access, inspections, traffic insights, and roadway safety. Federal guidance now directs agencies to connect AI adoption with mission effectiveness, governance, workforce capacity, transparency, and public trust. Public pension systems are likewise examining how boards should oversee technology and AI without becoming operational committees.^{[1][2][3]}

Those developments matter, but they do not answer the central institutional question:

THE INSTITUTIONAL QUESTION How can a public pension system turn AI possibility into durable public value without outrunning trust?

The question is intentionally larger than technology. AI touches member service, benefits administration, investments, actuarial work, finance, legal review, cybersecurity, procurement, records, workforce design, data governance, and public accountability. A local productivity tool can become an enterprise data pathway. A proof of concept can become an operational dependency. A vendor feature can quietly become part of a consequential decision.

The unit of analysis must therefore be the institution—not the model, tool, project, or department in isolation. The relevant system includes people, process, technology, data, incentives, governance, culture, vendors, risk, and mission. AI capability emerges when those elements can work together repeatedly, responsibly, and at the required level of performance.

NIST’s AI Risk Management Framework uses a lifecycle orientation—govern, map, measure, and manage—and its generative AI profile emphasizes adapting risk practices to goals, legal requirements, risk tolerance, and resources. GAO’s accountability framework similarly joins governance, data, performance, and monitoring. These standards reinforce a broader lesson: responsible AI is not a one-time approval. It is an institutional practice.^{[4][5]}

Why public pensions carry a distinctive stewardship burden

Public pension organizations combine long time horizons with immediate service obligations. They safeguard personal information, administer complex benefits, manage relationships with employers and members, support fiduciary investment processes, and operate under public scrutiny. Errors can affect individuals at consequential moments in their lives. Even when an AI use is not legally “high impact,” the institutional context may make it materially important.

This is why mission must come first. The point is not to slow useful innovation. It is to make innovation durable enough to deserve public reliance.

ACTIVITY IS NOT CAPABILITY

2. The False Momentum of Pilots

Pilots are valuable. They create a bounded space to learn, test assumptions, expose constraints, and build familiarity. The problem begins when leaders treat pilot completion as evidence of enterprise readiness.

A pilot often benefits from conditions that will not survive scale: hand-selected users, unusually attentive sponsors, manually cleaned data, temporary vendor support, exceptions to normal processes, narrow volumes, and limited consequences when something goes wrong. Those conditions can make an experiment look stronger than the institution around it.

A USEFUL DISTINCTION A pilot proves that something can work. Institutional capability proves that the organization can make it work repeatedly, responsibly, and without heroic intervention.

Five forms of false momentum

- **Pilot volume.** Many demonstrations create the appearance of progress while leaving ownership, integration, and operating capacity unresolved.
- **Local efficiency.** Time saved in a controlled test is treated as organizational value before quality, rework, adoption, supervision, and downstream effects are measured.
- **Success transfer.** A well-managed use case is assumed to establish readiness for other uses with different data, consequences, users, and failure modes.
- **Vendor velocity.** A vendor roadmap is mistaken for the institution's strategy, shifting decisions about data, dependency, and future operating models outside the organization.
- **Point-in-time governance.** Approval at launch is treated as durable assurance even though models, data, users, vendors, threats, and institutional conditions continue to change.

The practical implication is not to stop experimenting. It is to design experiments as evidence-producing investments. Before an experiment begins, leaders should know what uncertainty it is intended to reduce, what evidence would justify the next commitment, what conditions would stop it, and what institutional capabilities would be required if it succeeds.

A DIFFERENT FRAME

3. AI as Fiduciary Infrastructure

Public pension systems already understand portfolio thinking. They allocate scarce capital across competing opportunities, distinguish time horizons, evaluate risk and return, monitor changing conditions, and rebalance when evidence or objectives change. The same underlying discipline can improve technology decisions.

The analogy has limits. AI initiatives are not securities, and public value cannot be reduced to financial return. But the transferable structure is powerful: leaders must make scarcity visible, compare unlike

opportunities, understand dependencies, distinguish reversible experiments from consequential commitments, and allocate limited capital, talent, attention, data, and leadership capacity.

DANG PRINCIPLE Technology is fiduciary infrastructure. Technology investment is fiduciary capital.

This framing changes the conversation. The question is no longer, “Which AI ideas are exciting?” It becomes, “Which institutional outcomes deserve investment, what evidence supports the thesis, what risks and dependencies travel with it, and what must be true for the institution to sustain the result?”

From project list to living portfolio

A project list reports activity. A portfolio expresses choices. It shows what the institution is sustaining, protecting, improving, expanding, and transforming—and what it is choosing not to fund yet. It also reveals concentration risk: too many initiatives dependent on the same data, vendor, specialized workforce, integration platform, or executive sponsor.

Portfolio stewardship creates permission to stop. An experiment that does not produce sufficient evidence is not necessarily a failure. It may have purchased clarity at a reasonable cost. The greater failure is allowing sunk cost, enthusiasm, or reputation to turn an unresolved experiment into a permanent obligation.

THE PUBLIC LOGIC OF THE DANG DECISION TEST

4. Five Questions Before Scale

Scaling is not merely a technical decision. It is a commitment of institutional resources and reputation. Before making that commitment, boards and executives should be able to see a coherent answer across five lenses.

Lens	Question before scale	Evidence leaders should expect
Mission	Does the use materially advance institutional purpose or member value?	A defined public or organizational outcome; affected stakeholders; strategic alignment; a credible reason AI is appropriate.
Value	What measurable return should the investment create?	Baseline; intended benefit; cost and capacity implications; quality and service measures; time horizon; accountable owner.
Trust	How could the use strengthen or weaken confidence and responsible stewardship?	Reliability; transparency; fairness; accessibility; privacy; security; accountability; means of explanation and redress.
Risk	What could fail, who could be harmed, and what exposure would remain?	Failure modes; impact and likelihood; dependencies; legal and contractual review; human oversight; contingency and exit path.
Readiness	Can the institution operate and sustain the capability responsibly?	Data quality; process maturity; integration; workforce; governance; monitoring; vendor terms; ownership; resources.

TABLE 1 • These are public decision questions, not a proprietary scoring instrument.

The score is not the decision

A score can organize evidence. It cannot accept accountability. Leaders still need to understand evidence quality, critical conditions, residual risk, reversibility, institutional capacity, ownership, and rationale. The purpose of structure is to improve judgment—not to create the appearance that judgment is no longer necessary.

This is especially important when AI performance is probabilistic, use conditions are changing, or different stakeholders bear different benefits and risks. A single number can hide the very tradeoffs that governance must make visible.

SEE THE WHOLE SYSTEM

5. Build the Conditions for Capability

AI capability is not a product that can be purchased. It is an institutional condition that must be built. Current federal guidance reflects this systems view by connecting AI maturity with infrastructure, quality data, traceability, integration, security, privacy, governance, procurement, workforce literacy, and continuous monitoring.^[2]

Seven conditions that make scale responsible

- **Accountable leadership.** A named executive owns the outcome, and operating responsibility is clear across business, technology, data, security, legal, procurement, risk, and affected functions.
- **Process clarity.** The process being changed is understood, including exceptions, downstream effects, decision rights, records obligations, and the point at which human judgment must remain decisive.
- **Trusted data.** Source quality, provenance, access, retention, privacy, representativeness, and fitness for the intended decision are known well enough to support the use.
- **Adaptable architecture.** The organization can integrate, secure, monitor, update, and if necessary replace the capability without creating unacceptable fragility or lock-in.
- **Workforce capability.** Employees understand appropriate use, limitations, escalation, verification, and their continuing accountability for work supported by AI.
- **Proportional governance.** Oversight occurs when it can still improve the decision, and its intensity matches the materiality and reversibility of the use.
- **Measurement and learning.** The institution can observe both performance and consequences over time, learn from use, and act when the investment thesis no longer holds.

These conditions are not a reason to wait for perfection. No institution begins fully ready. They are a way to separate responsible learning from unmanaged dependence. A limited, reversible experiment may proceed with known gaps. A consequential, difficult-to-reverse deployment should require stronger evidence and stronger institutional conditions.

JUST-IN-TIME GOVERNANCE

6. Govern the Decision Moments

Governance often fails in one of two directions. It arrives too early and treats every idea as if it were a production system, or it arrives too late—after a vendor has been selected, data has moved, users have formed dependencies, and schedule pressure has narrowed the available choices.

Just-in-Time Governance places oversight at the moments when a decision changes the institution's exposure or commitment:

1. **Intent.** Clarify mission, use case, affected people, sponsor, and boundaries.
2. **Exploration.** Set safe conditions for learning, data use, access, evaluation, and documentation.
3. **Commitment.** Decide whether evidence justifies investment, procurement, integration, or broader reliance.
4. **Deployment.** Confirm ownership, controls, user readiness, monitoring, records, contingency, and communication.
5. **Scaling.** Evaluate whether institutional capacity and evidence support wider use or greater consequence.
6. **Reassessment.** Review performance, drift, incidents, stakeholder effects, vendor changes, and the continuing investment thesis.

The level of oversight should rise with member impact, data sensitivity, operational dependency, financial exposure, reputational risk, legal consequence, irreversibility, and institutional unreadiness. This mirrors the risk-proportionate direction in federal guidance and avoids building a separate bureaucracy around every use.^[2]

GOVERNANCE SHOULD ACCELERATE RESPONSIBLE INNOVATION. The purpose of governance is not to make decisions slower. It is to make consequential decisions clearer, earlier, and more accountable.

VALUE WITHOUT TRUST IS INCOMPLETE

7. Measure Dual Return

Many AI business cases stop at efficiency: minutes saved, transactions processed, documents summarized, or calls handled. Those measures matter, but they are not enough for a public institution.

An AI investment can create local productivity while weakening transparency, increasing rework, concentrating vendor dependence, confusing accountability, or reducing confidence in the institution. Conversely, a use that strengthens reliability, accessibility, explainability, or service consistency may create value that a narrow financial measure misses.

The appropriate standard is dual return:

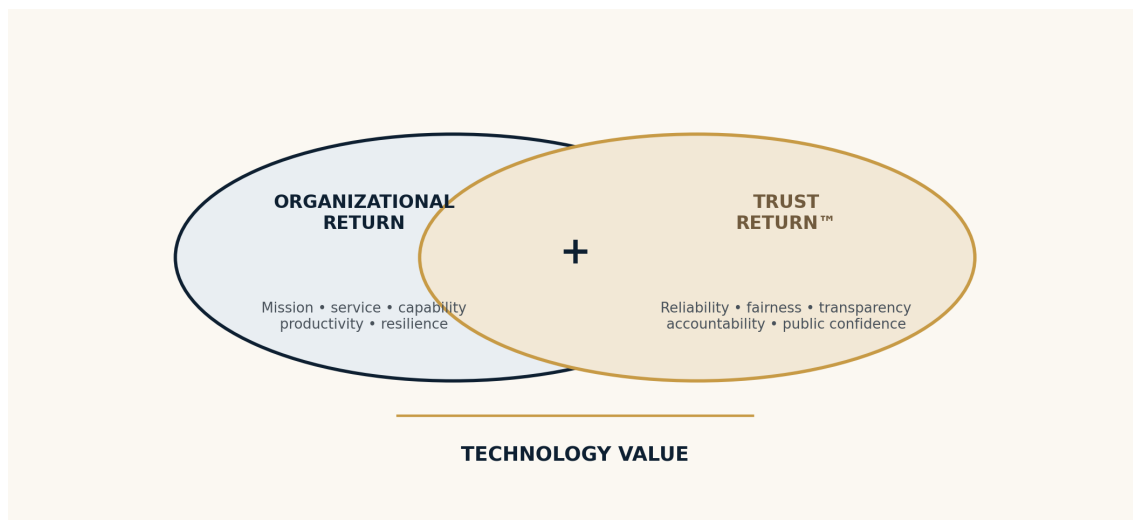


FIGURE 2 • *Technology Value = Organizational Return + Trust Return*

Organizational Return

Measure the outcomes the investment exists to create: member service, decision quality, accuracy, timeliness, workforce capacity, productivity, resilience, cost avoidance, financial value, compliance, or other mission-specific results. Compare performance with a credible baseline and include the cost of integration, supervision, verification, change, support, and exit—not only licensing.

Trust Return

Trust Return is the increase in confidence, credibility, transparency, fairness, reliability, accountability, responsible use, and stewardship generated by a decision or investment. It can be examined through reliability and service; security and privacy; transparency and explainability; fairness and accessibility; accountability and governance; data integrity; resilience; and public value.

Trust is not sentimental goodwill. It is an institutional outcome. It is built when people can rely on the service, understand appropriate parts of the decision, see who is accountable, challenge an error, and observe that the institution acts responsibly when evidence changes.

REBALANCE AS EVIDENCE CHANGES

8. Manage AI as a Living Portfolio

AI changes too quickly for a static roadmap to remain sufficient. Models improve. Costs move. Vendors revise terms. Laws and policies evolve. Threats emerge. Employees learn. Data conditions change. An institution that can approve AI but cannot reassess it has only half a governance system.

The portfolio should support five legitimate decisions: continue, condition, pause, stop, or scale. Each is a stewardship choice. Scaling is not the default reward for activity; stopping is not automatic evidence of failure.

A high-level stewardship cycle

ASSESS → ALIGN → PRIORITIZE → ALLOCATE → GOVERN → EXECUTE → MEASURE → REBALANCE The cycle keeps mission, capital, accountability, evidence, and learning connected.

The cycle also protects institutions from a common form of technical debt: obligations created by yesterday's enthusiasm that continue consuming tomorrow's capacity. Rebalancing makes room for stronger evidence, changing mission needs, and better options.

Boards do not need a dashboard of every experiment. They need a portfolio-level view of direction, concentration, material risk, expected value, trust implications, institutional readiness, and the decisions management is making as evidence changes.

GOVERN THE PORTFOLIO, NOT THE PROJECT PLAN

9. The Leadership Agenda

Public pension boards are actively working through the boundary between governance and operations. CalSTRS' 2026 board-level work emphasized clarifying where oversight is needed, when issues should come to the board, and how trustees can engage management through informed, forward-looking questions without being drawn into technical decisions.^[3]

Role	Primary responsibility	Questions that belong at this level
Board	Govern direction, material risk, value, trust, and institutional conditions.	Are strategy and risk tolerance clear? Which uses could materially affect members, fiduciary outcomes, public trust, or institutional resilience? What evidence demonstrates both returns?
Executive team	Align mission, investment, accountability, operating capacity, and enterprise priorities.	Who owns the outcome? What tradeoffs are being made? Are dependencies and readiness visible? What conditions trigger escalation, pause, or exit?
Technology and data leadership	Translate possibility into secure, integrated, measurable, and supportable capability.	Are architecture, data, monitoring, security, procurement, workforce, and lifecycle operations strong enough for the intended use?
Business and control functions	Define the process, human judgment, service outcome, constraints, and continuing accountability.	How does work actually change? Where must people verify, decide, explain, record, or intervene? What would harm look like in practice?

TABLE 2 • Clear boundaries allow boards to govern AI without becoming technical committees.

Board fluency is decision fluency

Trustees do not need to become model engineers. They need enough fluency to understand institutional consequence: what AI is being asked to do, who is affected, what evidence supports reliance, how human accountability is preserved, what concentration or dependency is being created, and how the institution will know when the use should change.

BUILD THE CAPACITY TO KEEP DECIDING

10. A Practical Starting Agenda

Institutions at different maturity levels should not copy the same roadmap. But every public pension system can begin with a small number of leadership moves that improve the next decision without requiring a new bureaucracy.

1. **Name the institutional outcomes.** Identify the member, mission, service, risk, workforce, or decision outcomes AI might materially improve. Separate desired outcomes from attractive tools.
2. **Build a visible use-case inventory.** Include sanctioned experiments, embedded vendor features, employee-created workflows, procured capabilities, and consequential uses. Visibility is the beginning of stewardship.
3. **Select a small portfolio of learning investments.** Choose uses that matter, can be bounded, and reduce different forms of uncertainty. Avoid funding many similar demonstrations that produce the same lesson.
4. **Define evidence before enthusiasm takes over.** State the baseline, intended return, trust implications, risk, readiness gaps, stop conditions, and next decision before the experiment begins.

5. **Put governance at the decision moments.** Clarify who can authorize exploration, commitment, deployment, scaling, and continued use—and what conditions require broader review.
6. **Prepare the workforce for accountable use.** Teach employees not only how to prompt or automate, but how to verify, protect information, preserve records, recognize limits, escalate concerns, and remain responsible for the work.
7. **Review the portfolio on a regular cadence.** Continue, condition, pause, stop, or scale based on evidence. Capture what the institution learned so knowledge survives beyond a project or individual.

THE PRACTICAL TEST Philosophy becomes practice when it changes the next decision.

The goal is not to build a perfect AI governance system before anyone learns. The goal is to create a disciplined loop in which learning improves governance, governance improves decisions, decisions build capability, and capability earns trust.

STEWARDSHIP BEFORE SPECTACLE

Closing: The Institution Is the Enduring Advantage

The most important AI advantage available to a public pension system is not early access to a model. Models will change. Vendors will change. The language of the market will change.

The enduring advantage is an institution capable of learning, choosing, governing, executing, measuring, and adapting with purpose. Such an institution can use today's technology without becoming captive to it. It can move with appropriate speed without confusing speed with wisdom. It can innovate without asking members and the public to finance unmanaged uncertainty with their trust.

Technology changes. Principles endure.

The standard is not whether technology was delivered. The standard is whether the institution became stronger.

NORTH STAR Make technology a force that elevates people and mission.

Appendix: Questions for the Next Board Conversation

These questions are designed to improve strategic oversight. They are not a compliance checklist and should be adapted to the institution's authority, governance model, risk tolerance, and current maturity.

- **Mission.** Which member, fiduciary, service, workforce, or institutional outcomes are important enough to justify AI investment?
- **Portfolio.** Where are we concentrating money, attention, data, vendor dependency, or specialized talent—and what are we choosing not to fund?
- **Evidence.** What uncertainty is each material experiment intended to reduce, and what evidence would justify the next commitment?
- **Trust.** How could the use affect reliability, fairness, accessibility, transparency, accountability, privacy, security, and public confidence?
- **Readiness.** Which institutional conditions are strong enough today, which must be built, and which gaps are acceptable only because the use is bounded and reversible?
- **Governance.** At which decision moments should the board, executive team, or control functions become involved?
- **Workforce.** How will employees learn to use AI responsibly while remaining accountable for judgment, service, records, and outcomes?
- **Vendors.** What rights, dependencies, performance obligations, data protections, monitoring access, and exit options are being created?
- **Measurement.** How will we know whether the investment created Organizational Return and Trust Return?
- **Rebalancing.** What conditions would cause us to continue, condition, pause, stop, or scale—and how often will we revisit the thesis?

AUTHORITATIVE PUBLIC REFERENCES

Sources and Further Reading

1. National Conference on Public Employee Retirement Systems (NCPERS), “From Experimentation to Enterprise: Building an AI Strategy for Public Pension Systems,” July 30, 2026. [View source](#)
2. Office of Management and Budget, Memorandum M-25-21, “Accelerating Federal Use of AI through Innovation, Governance, and Public Trust,” April 3, 2025. [View source](#)
3. California State Teachers’ Retirement System, CEO Report: Enterprise Technology Governance and Artificial Intelligence Oversight, March 5, 2026. [View source](#)
4. National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework (AI RMF 1.0) and associated resources. [View source](#)
5. U.S. Government Accountability Office, “Artificial Intelligence: An Accountability Framework for Federal Agencies and Other Entities,” GAO-21-519SP, June 30, 2021. [View source](#)
6. National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile,” NIST AI 600-1, July 2024. [View source](#)
7. State of California, Generative Artificial Intelligence portal and public-sector use cases. [View source](#)

How these sources are used

The public standards above provide context and corroboration.

LEAD WITH PURPOSE.

Govern with confidence.
Invest in the future.
Earn public trust.